CS246: Database Management Systems Lab

Week # 04 (1 Questions, 92 Marks)
Lab session: AL1
Held on: 02-Feb-2026 (Mon)
Lab Timings: 14:00 to 17:00 Hours   Pages: 3
Submission dead line: 17:00 hrs, 02-Feb-2026
Submissions accepted from: 16:45 hrs, 02-Feb-2026
Submissions accepted till: 17:30 hrs, 02-Feb-2026
Every one minute delay in submission will result in one mark deduction
Instructors   Dr. V. Vijaya Saradhi
Head TA   Divanshu Walia
Department of CSE, IIT Guwahati

Question 1: (92 points)

This week's lab focuses on relational algebraic expressions using the online tool `https://dbis-uibk.github.io/relax/`.

You will express core computational tasks from:

- Week 02: ATM locations and
- Week 03: Flight delays

Using relational algebra.

ATMs Contains the following attributes

1. bank: string
2. lat: number
3. lon: number
4. atm_id: string

Users Contains the following attributes

1. user_id: string
2. bank: string
3. lat: number
4. lon: number

Flights Contains the following attributes

1. flight_id: number
2. origin: string
3. dest: string
4. sched_year: number
5. sched_month: number
6. sched_day: number
7. sched_hour: number

8. sched_min: number
9. act_year: number
10. act_month: number
11. act_day: number
12. act_hour: number
13. act_min: number

Airports Contains the following attributes

1. code: string
2. city: string
3. state: string

Routes Contains the following attributes

1. origin: string
2. dest: string
3. distance_km: number

Instructions Use the RelaX tool to define these relations. This tool allows extended RA for computations.

Given two (lat_1, lon_1) and (lat_2, lon_2), the distance formula is: $dist = 111 \times \sqrt{(lat_2 - lat_1)^2 + (lon_2 - lon}$

Use of the function $\sqrt{()}$ in relational algebraic expressions is allowed in this tool.

The tutorial slides corresponding to this tool are shared with this assignment.

Write relational algebra expressions for each task.

Task 01 (4 marks) Upload the database `week04.csv` into the tool. Make sure that this tool is using the week04 database. To achieve this,

1. Click on the tab `Group Editor`
2. Empty the space in this editor.
3. Copy and Paste contents of the file week04.csv.
4. Press the button "Preview"
5. Click the link "use Group in editor" link. This step is crucial for loading the custom data into the tool

Task 02 (4 marks) Project the bank and atm_id from ATMs.

Task 03 (4 marks) Select ATMs where bank = 'SBI' from ATMs.

Task 04 (4 marks) Rename the relation ATMs to BankATMs (no column changes).

Task 05 (4 marks) Rename the column atm_id to atm_identifier in ATMs.

Task 06 (4 marks) Rename lat to lat1 and lon1 to longitude in ATMs.

Task 07 (4 marks) Rename columns bank to bank_name, atm_id to atm_code, and rename relation AMTs to ATM_Locations.

Task 08 (4 marks) Compute the $\sqrt{(lon - lat)}$ from Users table.

Task 09 (12 marks) For each origin airport, calculate the average delay in minutes across all flights departing from that airport. The resulting relation should have the following columns: origin and computed average delay_minutes.

Use only act_min and sched_min for computing the delay information.

Task 10 (16 marks) Calculate the average delay of flights for each departure airport (origin), then show only those airports where the average delay is less than 20 minutes, and display the airport code along with that average delay value.

Task 11 (32 marks) For each user, list all ATMs of any bank that are within 1 km of that user's position.

Submission Instructions Write all the queries in the same editor.

1. Download all the relational algebra queries data into a text file named after your roll number.
2. For each query, download the result in a CSV file. Name the result files as follows: `roll-number-task-number.csv`.
3. For a student with roll number: `240101999` solved Task 02 the the result output should be saved into the file named as: `240101999-task02.csv`
4. For a student with roll number: `240101999` solved Task 03 the the result output should be saved into the file named as: `240101999-task03.csv`
5. If you are confused with the file naming, ask only the following TAs: For CSE students: Sanyam Garg
   For M&C students: Rishika Bhattacharyya
6. If the output files are named properly you will receive ZERO marks for the corresponding task. No further queries will be entertained on the file naming convention.
7. Place the query file and output of each task into a zip file named after your roll number. For a student with roll number `240101999` the zip file name is: `240101999.zip`. You should upload this zip file in MS Assignments.